

SOUTH CENTRAL RLY

**HYDERABAD
DIVISION-S AND T
2nd Floor, S and T
Department,
Hyderabad Bhavan,
Secunderabad.
Secunderabad,
500017
Telangana, India**

**Letter No: HYDERABAD DIVISION-S AND T /
Y-SG-36-2024-25-03 / 01052610112449**

**Dated:
10/09/2024**

**M/s NAGALAKSHMI ENTERPRISES-
SECUNDERABAD**

**PLOT No 172 RAVI COLONY MAHENDRA
HILLS EAST MARREDPALLY
SECUNDERABAD
SECUNDERABAD- 500026
Telangana, India**

Sub: Letter Of Acceptance

- Ref: 1.** Tender No. Y-SG-36-2024-25-03 closing date 10-07-2024 11:30 for Hyderabad Division -: Signal & Telecom arrangements in connection with provision of alternate Control office Setup of Hyderabad Division (2) STTC/MLY:- Provision of CCTV cameras for STTC/MLY campus. (3) Provision of Route setting and Non route setting panels Stations (indoor works) at STTC / MLY (4) SC-DHNE Sec - Elimination of Manned Level Crossing Gate No. 77 at Km No. 126/500-600 by providing RUB between MBNR-MQN Stations - Signal & Telecom arrangements (5) SC-DHNE Section - Elimination of Manned Level Crossing Gate No. 92 at Km No. 169/0-100 by providing RUB between WPR-SRNR Station - Signal & Telecom arrangements (6) SC-DHNE Section - Elimination of Manned Level Crossing Gate No. 96 at Km No. 180/0-100 by providing RUB between SRNR-GWD Stations - Signal & Telecom arrangements (7) SC-MUE Section - Elimination of Manned Level Crossing Gate No. 207 at Km 504/400-500 by providing RUB between UPW-KMC Stations - Signal & Telecom arrangements (8) SC-MUE Section - Elimination of Manned Level Crossing Gate No. 216 at Km 528/700-800 by providing RUB between TMD-BKU Stations - Signal & Telecom arrangements (9) SC-MUE Section - Elimination of Manned Level

Crossing Gate No. 225 at Km 550/300-400 by providing RUB between MZL-WDR Stations - Signal & Telecom arrangements (10) SC-DHNE Section :--Falaknuma- Augmentation of traction power supply infrastructure over HYB Division. - Telecom arrangements (11) Provision of Telecom arrangements in connection with the work of Improvements to Existing Simulator room for Accommodation of Women RPF Personal including Furniture at RPF/TC/MLY & (12) Replacement of Codal Life completed SDH equipment in SC-MUE, SC-DHNE & HQrs Area in Hyderabad Division.

2. Your bid ID **16958869** dated **10/07/2024 10:53**
3. Your Negotiation bid IDs [**17247452** dated **05/09/2024 16:24**]

The Competent Authority has accepted your offered rates in connection with the subject work. The total cost of the work at the accepted rates works out to Rs. 30529885.42 (Rupees Three Crore Five Lakh Twenty-Nine Thousand Eight Hundred And Eighty-Five Rupees And Fourty-Two Paise Only)

A sum of Rs.273700 deposited as Earnest Money vide IREPS reference ID PE196022942912 has been retained towards initial Security Money for due and faithful fulfillment of the contract, and the balance Security Money will be recovered from the progressive bills @ 6 % of the bill amount till it reaches 5 % of the contract value in terms of clause 16(1) of GCC.

You are requested to submit Performance Guarantee in the form as given in Clause 16(4) of GCC equivalent to 5 % of the contract value amounting to Rs. 1526494.27 (Rupees Fifteen Lakh Twenty-Six Thousand Four Hundred And Ninety-Four Rupees And Twenty-Seven Paise Only) within 21 days from the date of issue of Letter of Acceptance, valid up to stipulated date of completion plus 60 days so that contract agreement can be executed.

The entire work shall be completed within 15 month from the date of issue of Letter of Acceptance.

Performance Guarantee (PG) : The successful bidder shall have to submit a Performance Guarantee (PG) within 21 (Twenty one) days from the date of issue of Letter of Acceptance (LOA). Extension of time for submission of PG beyond 21 (Twenty one) days and upto 60 days from the date of issue of LOA may be given by the Authority who is competent to sign the contract agreement. However, a penal interest of 12% per annum shall be charged for the delay beyond 21(Twenty one) days, i.e. from 22nd day after the date of issue of LOA. Further, if the 60th day happens to be a declared holiday in the concerned office of the Railway, submission of PG can be accepted on the next working day.

a): In all other cases, if the Contractor fails to submit the requisite PG even after 60 days from the date of issue of LOA, the contract is liable to be terminated. In case contract is terminated railway shall be entitled to forfeit Earnest Money Deposit and other dues payable against that contract. In case a tenderer has not submitted Earnest Money Deposit on the strength of their registration as a Startup recognized by Department of Industrial Policy and Promotion (DIPP) under Ministry of Commerce and Industry, DIPP shall be informed to this effect. The failed Contractor shall be debarred from participating in re-tender for that work. The format for submission of Bank Guarantee is enclosed. This PG shall be initially valid up to the stipulated date of completion plus 60 days beyond that.

b): Performance Guarantee to be deposited in favor of SENIOR DIVISIONAL FINANCE MANAGER/ HYDERABAD DIVISION, SOUTH CENTRAL RAILWAY, SECUNDERABAD.

1) Section of work: HQ Area & SC-MUE & SC-DHNE Section

2) Officer In charge: ADSTE/HYB, ADSTE/North/HYB & ADSTE/South/HYB

3) Signalling Supervisor in charge: SSE/Sig/GWD, SSE/Sig/MBNR, SSE/Sig/MED & SSE/Sig/SW

4) Telecom Supervisor in charge: SSE/Tele/CTO/HYB, SSE/Tele/LGD, SSE/Tele/KCG, SSE/Tele/GWD, SSE/Tele/MBNR, SSE/Tele/MJF & SSE/Tele/NZB

5) Stores-receipts/issues: SSE/Sig/CKL & SSE/Tele/Stores/LGD

6) Materials inspected by Receipts/issues: Officer: ADSTE/HYB & ADSTE/South/HYB

7) Materials inspected by Receipts/issues: Supervisor: SSE/Sig/CKL & SSE/Tele/Stores/LGD

8) Recording of measurements for : Supply Portion for Telecom Items: Officer: ADSTE/HYB

9) Recording of measurements for : Supply Portion for Telecom Items: Supervisor: SSE/Tele/Stores/LGD

10) Recording of measurements for : Supply Portion for Signal Items: Officer: ADSTE/South/HYB

11) Recording of measurements for : Supply Portion for Signal Items: Supervisor: SSE/Sig/CKL

12)Recording of measurements : for Labour Portion Officer:
ADSTE/HYB, ADSTE/North/HYB & ADSTE/South/HYB

13) Recording of measurements : for Labour Portion for Telecom
items : Supervisor: SSE/Tele/CTO/HYB, SSE/Tele/LGD,
SSE/Tele/KCG, SSE/Tele/GWD, SSE/Tele/MBNR, SSE/Tele/MJF
& SSE/Tele/NZB

14) Recording of measurements: for Labour Portion for Signal
items : Supervisor: SSE/Sig/GWD, SSE/Sig/MBNR,
SSE/Sig/MED & SSE/Sig/SW

15) Bill paying authority: Sr.DFM / HYB

All Other terms and conditions, as stipulated in the tender
documents shall be applicable.

TAMADA NEELA PAVANI

Sr.DSTEHYB

Digitally Signed

[View Signature Details](#)

Awarded Quantities And Rates

Item Sno.	Item Desc	Item Code	Item Qty	Qty Unit	Unit Rate(Rs)	Escl. (%)	Advt.Value (Rs)	Bid Rate/ Unit Rate	Bid Amount (Rs)
Schedule A-Schedule - A (Item Directory - Not Applicable)							2616957.00		
1	Supply Portion SOR Items		View Details		At Par		2616957.00	45.00 % Above	3794587.65
Schedule Totals							3794587.65		
Schedule B-Schedule B (Item Directory - Not Applicable)							2622317.39		
2	Labour Portion SOR Items		View Details		At Par		2622317.39	45.00 % Above	3802360.22
Schedule Totals							3802360.22		
Schedule C-Schedule C (Item Directory - Not Applicable)							2610044.50		
3	Supply Portion Non SOR Items		View Details		At Par		2610044.50	26.19 % Above	3293615.15
Schedule Totals							3293615.15		
Schedule D-Schedule D (Item Directory - Not Applicable)							7153218.48		
4	Labour Portion Non SOR Items		View Details		At Par		7153218.48	26.19 % Above	9026646.40

Schedule Totals				9026646.40			
Schedule E-Schedule E (Item Directory - Not Applicable)				9736400.00			
5	SUPPLY PORTION (STMs)	View Details	At Par	9736400.00	9.00	% Above	10612676.00
Schedule Totals				10612676.00			
Total Value				24738937.37			30529885.42
				Rebate on Total Value (%)		0.00	
				Net Bid Value		30529885.42	

Item Breakup

Schedule	A-Schedule - A					
Item - 1	Supply Portion SOR Items					
S No.	Item No	Description of Item	Unit	Qty	Rate	Amount
1	1	Supply of HDPE telecom duct with outer diameter of 40mm and inner diameter of 33mm permanently lubricated as per Specn. No. RDSO/SPN/TC/45/2006 Revision 1.0 or latest or as per specification TEC GR No.G/CDS-08/02,NOV 2004 or latest . HDPE pipe shall be supplied along with 4mm dia polypropylene rope of approved quality inside wherever required. The length of HDPE telecom duct should be 1 KM. End plugs to be provided along with drum (free of cost) (2 No. per Km.).	Kilometre	23	45226.00	1040198.00
2	2	Supply of HDPE duct associated accessories and tools as per the specification of NS.1: (i) Plastic couplers - Slip fit / push type for HDPE duct (2 No. per Km.) ii) End plug for HDPE duct (2 No. per KM). iii) Cable sealing plugs for HDPE duct (1 No. of per Km.)	Numbers	21	110.00	2310.00
3	3	Supply of WAGO Terminal as per RDSO Specification No. 189/2004 which consists as per following: a) 2 Conductor disconnect type terminal block as per Model No. 280-870 -- Qty: 6 No. b) End plate type 280-374 suitable for disconnect terminal of Model No.280-870 -- Qty: 1 No. c) End stop terminal block type 249-117 -- Qty: 2 No.	Set	450	486.00	218700.00

		d) Shorting links (16A) as per model 280-402.				
4	4	Supply of Markers for the WAGO terminals as per Model No. 249-502 (Plain masker) - 100 No.	Set	45	43.00	1935.00
5	5	Supply of hylam sheet, resin bonded, 6mm thick, grade P3, to IS specification IS: 2036 of 1974 or latest and size 2400mm X 1200mm of colour approved by the site in-charge	Numbers	5	4821.00	24105.00
6	6	Supply of Relay rack universal type as per drg.no.S&T/MFT/291 with scaffolding, POWDER COATED with stainless steel nuts and bolts for fixing the racks. Supporting angles, frame mounting triangle base with 'J' bolts and insulation of required numbers complete to suit QN1 / K-50 pre wired tag blocks with suitable inner frames. - 2 way	Numbers	5	10278.00	51390.00
7	7	Supply of Cable termination racks size 880mm x 2 Meters made out of 50mmx50mmx6mm thick MS angle with stainless steel nuts and bolts for fixing of the rack with POWDER COATED and drilling suitable holes to suit hylam sheet and relay frames complete as per standard practice with tripod bases and reel insulators (for K rack, P rack and power rack) and with cable ladder of width 400mm and length of 2 Mtr.. The thickness of angle to be 5mm. two angles of 40mmX40mmX5mm to be supplied at both ends and supporting flats of 25mm X5mm.	Numbers	3	5125.00	15375.00
8	8	Supply of terminal block 1 way PBT 60mm as per RDSO Drg.No. SA.23745/Alt.5 & IRS S.75/2006 (Rev.2) & IRS.S.23 with latest amendment.	Numbers	100	61.00	6100.00
9	9	Supply of terminal block 6 way PBT 25mm as per RDSO Drg.No. SA.23756/Alt.3 & IRS S.75/2006 (Rev.2) & IRS.S.23 or latest.	Numbers	170	285.00	48450.00
10	10	Supply of Non-deteriorating type of fuse holders 20A capacity - for Railway signalling Rated (in Rupees) voltage 240 AC/DC to Specn. No. IRS S.78/92 (IRS-S-78/97 Tentative) IS:13703 (part	Numbers	320	150.00	48000.00

		2/sec2):1993 suitable for E1 fuse 20Amps SC-20 Holder Safe Clip / knife edge type size E1 (R-SS) amp.Fuse holder 20A capacity				
11	11	Supply of Non-deteriorating type of fuse links 2 Amps- electric fuse for Railway signalling Rated (in Rupees) voltage 240 AC/DC to Specn. No. IRS S.78/92 (IRS-S-78/97 Tentative) IS:13703 (part 2/sec2):1993 suitable for SC-20 Holder Safe Clip / knife edge type	Numbers	200	20.00	4000.00
12	12	Supply of Non-deteriorating type of fuse links 16 Amps- electric fuse for Railway signalling Rated (in Rupees) voltage 240 AC/DC to Specn. No. IRS S.78/92 (IRS-S-78/97 Tentative) IS:13703 (part 2/sec2):1993 suitable for SC-20 Holder Safe Clip / knife edge type	Numbers	100	21.00	2100.00
13	13	Supply of Non-deteriorating type of fuse links 10 Amps- electric fuse for Railway signalling Rated (in Rupees) voltage 240 AC/DC to Specn. No. IRS S.78/92 (IRS-S-78/97 Tentative) IS:13703 (part 2/sec2):1993 suitable for SC-20 Holder Safe Clip / knife edge type	Numbers	100	21.00	2100.00
14	14	Supply of Non-deteriorating type of fuse links 6 Amps- electric fuse for Railway signalling Rated (in Rupees) voltage 240 AC/DC to Specn. No. IRS S.78/92 (IRS-S-78/97 Tentative) IS:13703 (part 2/sec2):1993 suitable for SC-20 Holder Safe Clip / knife edge type	Numbers	100	21.00	2100.00
15	15	Supply of PVC insulated wire copper conductor 16/02mm to ATC IS:694 to RDSO specification no.IRS.S76/89 Amendment.3 or latest. Each coil of 100 mts length. (To be procured from RDSO approved firms.) (Unit : Coil)	Numbers	300	797.00	239100.00

16	16	Supply of PVC insulated wire multi-strand copper conductor 10 Sq.mm flexible Specn.IRS.S76/89. Amendment No.3 or latest. Each coil of 100 mts length. To be procured from RDSO approved firms) (size 140/0.3mm) (Unit : Coil)	Numbers	6	9390.00	56340.00
17	17	Supply of PVC wire coils of size 28/03 multi strand (flexible). Specn.IRS. S76/89. Amendment No.3 or latest. Each coil of 100 mts length. (To be procured from RDSO approved firms). (Unit: Coil)	Numbers	8	1635.00	13080.00
18	18	Supply of Four channel Earth leakage detectors as per RDSO specification No. 256/2008 or latest to the required configuartion as per site incharge directions	Numbers	1	53628.00	53628.00
19	19	Supply of Transparent plastic cover superior quality for keeping circuit diagram sheets with provision for hanging size "C".	Numbers	100	40.00	4000.00
20	20	Supply of various type of lugs as per instructions of site in charge	Numbers	400	4.00	1600.00
21	21	Supply of PVC conduit pipe 3 Mtr. length 25mm dia with bends etc as per site requirement Make: Nandi/Sudhakar or similar	Numbers	4	35.00	140.00
22	22	Supply of copper lugs10 Sqmm of reputed make	Lumpsum	400	8.00	3200.00
23	23	Supply of Copper eyelets to suit 16/0.2 wire Dowell's make	Numbers	1500	2.00	3000.00
24	24	Supply of Copper eyelets to suit 10.sqmm wire Dowell's make	Numbers	400	4.00	1600.00
25	25	Supply of Domino type operation cum indication panel 3/4 road with LED indicators for tracks, signals, crank handles etc., and knobs/push button for points and signal operation for as per the panel diagram supplied by the railways. The layout shall be painted with screen printing. Following spares to be supplied with each Panel. 1. Knob/push buttons -- 2 No. 2. Domino with cancellation button -- 1 No. 3. Spare Domino of different type - - 5 No. 4. 24V Impulse counter - 2 No. 5. Reminder caps - 10 No.	Numbers	2	121676.00	243352.00

		NOTE :- Prior to transportation of panel to site, the same should be supplied with internal wiring done from domino to terminal block neatly bunched. The panel wiring particulars should be submitted along with panel at factory premises for inspection. Knobs / push buttons to be procured from approved firms.				
26	26	Supply of splice enclosure (24 Fiber) to RDSO spec No RDSO/SPN/ TC/68/ 2006 Rev.o.Amdt 1 or latest (To be procured from RDSO approved firms).	Numbers	68	2417.00	164356.00
27	27	Supply of Thermo shrinkable jointing kit as per Spec. No. IRS TC. 77/2012 (Rev 3.0) Amendment.1 to 3 or latest for all types joints suitable for 6 Quad Jelly filled UG cable.(To be procured from RDSO approved firms).	Numbers	62	1923.00	119226.00
28	28	Supply of GKP type Location Boxes Full size as per Drg No. S&T/MFT/ 2378 (Single)	Numbers	12	8960.00	107520.00
29	29	Supply of 'E' type locks of different wards as per requirement with key per E-type lock to Drg. No. SA.3376/M/Latest and RDSO Specn. No. IRS. S.30/64.	Numbers	12	560.00	6720.00
30	30	Supply of basic material to construct unit maintenance free earth consisting as per RDSO spec no.RDSO/SPN/197 ver.1.0 or latest consisting of (To be Procured from RDSO approved firm) as per bonding connections drawing no.SDO/RDSO/E&B/002. a) copper bonded steel electrode of 3.0m (min) long, 17.0 mm dia (min) with copper bonding thickness mini 250 micron and UL listed and marked--1 no. b) Earth enhancement compound of RDSO approved brand supplied in sealed bags of min 30-35 Kgs c) Copper strip of 200X25x6mm to exothermically welded to earth electrode-1 No. d) Copper strip of 300X25x6mm (MEEB)-1 No. e) Copper strip of 150X25x6mm (SEEB)-1 nos f)	Set	2	16837.00	33674.00

		Copper Tape of 25x2mm or 8mm dia-1 nos				
31	31	Supply of 10 Sq mm Multi strand single core PVC insulated copper cable as per IS:694 for connecting various equipments to SEEB.(To be procured from RDSO approved firms).	Metre	50	87.00	4350.00
32	32	Supply of 35 Sq.mm Multi-strand single core PVC insulated copper cable as per IS:694 for connecting Main earth electrode to MEEB in the equipment room in duplicate.(To be procured from RDSO approved firms).	Metre	160	303.00	48480.00
33	33	Supply of 16 Sq mm Multi-stand single core PVC Insulated copper cable as per IS:694 for connecting MEEB to SEEB and SPDs and SPDs to MEEB.(To be procured from RDSO approved firms).	Metre	150	144.00	21600.00
34	34	Supply of desk type 2 wire , 12 way DTMF selective calling group telephone complete as per specification IRS. TC. 80-2000 or latest with suitable power supply unit.(To be procured from RDSO approved firms).	Set	4	6282.00	25128.00
					Total	2616957.00
Schedule	B-Schedule B					
Item - 2	Labour Portion SOR Items					
S No.	Item No	Description of Item	Unit	Qty	Rate	Amount
1	1	Laying different type cables in the trench. Termination covered elsewhere.	RMT	32200	5.00	161000.00
2	2	Laying of HDPE duct 40/33 mm dia in the trenches and in protective works already provided.	Kilometre	5	4956.00	24780.00
3	3	Blowing of armoured optic fiber cable (OFC Cable supplied by Railways) into the duct supplied vide NS.2 of supply portion by using blowing machine and associated works. Before blowing the duct integrity shall be tested. If any defect is found during duct integrity test, the entire length of HDPE duct from one chamber to another	Kilometre	23	7382.00	169786.00

		chamber is to be replaced by the contractor at free of cost.				
4	4	Excavation of Jointing pit and provision of joint chamber of 1.2Meter dia. Of 60 cm height with 2-piece cover on top and bottom cover with holes for drainage (reinforcement drawing will be as per technical specification).This includes water cooling of chamber, backfilling and ramming of the trench after cable laying and consolidation of soil as well as disposal of soil as per plan as advised by site engineer.	Numbers	24	3745.00	89880.00
5	5	Excavation of trench to a depth of 1200mm and width of 300m at bottom in all types of soil(Normal soil/soft soil/sandy soil except in hard morum / rocky soils /bridges/ level crossings/ track crossings/ coil pits,) as per RDSO Sketch no SDO/CABLE LAYING /003 including marking of cable alignment, clearing of jungle, bushes, trial pits if required, refilling with excavated soil, ramming and consolidation after laying of cables complete to the finished item of work as directed by supervisor in-charge.	Route Kilo Meter (RKM)	16.2	51912.00	840974.40
6	6	Installation of OFC joint enclosure and splicing of Optic fiber cable (24 fibres) and testing as per technical specification No. GR OJC-2/02 Sept 2003. Supply of Splice Enclosure (24 fiber) shall be of TVS make Part No.1500-L18X-XX as per Board specification of TEC covered in Schedule of supply). All materials including Fusion splicing machine used for fiber splicing and, OTDR for testing and other testing materials if any shall be arranged by the contractor	Numbers	32	3999.00	127968.00
7	7	Breaking of metal/ tar road and refilling after laying the cables as per drawing no.SDO/CABLE LAYING/010. The surface of the road shall be concreted to match with the old surface	RMT	55	142.00	7810.00
8	8	Transporting and supply and installation of Concrete (RCC)	Numbers	295	272.00	80240.00

		route indicators at intervals of 50 meters as per technical specification. They should be of standard RCC with letters "SCR S&T cable" as per type of cable laid to drawing . In case of telecom quad cable joint locations it shall be painted with red color.				
9	9	Supply & Laying of G.I Pipes 50mm dia. medium quality to specification IS.1239 (part-I) 1990 and galvanised to specification IS.4736-1986 On bridges , culverts, trench with channel ramps on either end of the culvert/bridges. Clamping arrangement with MS brackets for fixing of GI pipes on bridges/culverts, small holes of 5mm are to be drilled on the pipe before laying of the pipes.	RMT	50	254.00	12700.00
10	10	Supply and installation of rack mounted 12F FMS fiber part no.fms-1u 12FS 12FP Raychem make including 12Nos 0.2db SC /PC adaptors and 12 numbers 900 micron 1.5 meter FCPC pigtails.RDSO Spec No.: RDSO/SPN/TC-037-2000 (Ver.3) or latest. Make: Rachem, Rosenberger, Bestnet	Numbers	11	9266.00	101926.00
11	11	Supply and installation of rack mounted 24F FMS fiber part no.fms-1u 24FS 24FP including 24Nos 0.2db SC/PC adaptors and 24numbers 900 micron 1.5 meter SCPC pigtails. RDSO Spec No.: RDSO/ SPN/TC-037-2000 (Ver.3) or latest. Make: Rachem, Rosenberger, Bestnet	Numbers	8	15443.00	123544.00
12	12	Routing and connecting of all the SPDs in the equipment room to MEEB with 16 sq mm copper cable using copper lugs with stainless steel nut and bolts (for item procured against supply).	RMT	150	23.00	3450.00
13	13	Routing and connecting all the equipments in the equipment/relay room to MEEB/SEEB with 10 sq mm copper cable using copper lugs with stainless steel nuts and bolts and fixtures. The work includes laying and routing the copper cable in PVC	RMT	150	23.00	3450.00

		pipes/conduits and fixing them on wall.				
14	14	Supply and installation of 45mmx 45mm PVC channels for drawing switchboard cables at OFC cable huts and at stations as indicated by Railway Engineer at site.	Metre	630	66.00	41580.00
15	15	Cable termination including fixing terminals, fuses, LDs etc., Meggering the cables and recording the readings.	Per Conductor	1399	7.00	9793.00
16	16	Fixing relay bases, wiring in location boxes, bunching, fixing of relays as per circuit diagram. This includes fixing of condensers, resistance, 6 way / 1 way terminals/ WAGO, ND type fuses, LED indicators for fuses as per type fuses, LED indicators for fuses.	Per Relay	20	156.00	3120.00
17	17	Providing crank handle box, wiring, testing as per instructions of site incharge and including concealed wiring in the wall .	Numbers	4	3848.00	15392.00
18	18	Installation, testing and commissioning of Data Logger which includes the software configuration needed, validation, taking test print, for networking for transferring the data to nominated location for monitoring and changing of EPROMS if any as per the instructions of the site in-charge, including alterations data base in Hq/SSE offices where ever required.	Numbers	1	24280.00	24280.00
19	19	Erection of Relay rack 1way/2way/4way with scaffolding suitably concreting tri-pod bases using J-bolts, fixing rack supporting of size 50x50x6 mm on to the wall. Reel insulators are to be provided for insulating the rack from the scaffolding & tri-pod bases. (payment based on number of equivalent 1 way relay racks). (Unit : Per one way rack)	Each	16	341.00	5456.00
20	20	Fabrication and supply of MS frames to suit 'Q' series relays to accommodate 6 No. in each set as per Drg.No. B.SG.CN.	Numbers	160	294.00	47040.00

		MISC.9.98 duly drilling required holes for relay bases (All materials such as frames, bolts, nuts and washers to be supplied by the contractor) (Only Relay bases will be supplied by the Railways).				
21	21	Fixing relay bases on the relay frames erected in Relay Room, Wiring of all types of relays by drawing various sizes of wires/multicore cable, fixing fuses, condensers, resistances, LEDs etc and wiring duly soldering the same as per approved circuit diagram, testing point to point before soldering and after soldering, bunching and lacing as per detailed technical specification. Different colour codes and sizes of wire to be used as per standard practice. This also includes, provision of all PVC sleeves required to be provided. Copper lugs to suit to 16/02 wire and neat bunching, lacing with Modi Nylon / Twine thread and using PVC dotted tape with buttons were required. This also includes soldering all the wires in the relay room with Rosine core of IS Spec. 1921 @40<37@ Tin and 40% Led). Soldering should be done as per standard practice. This also includes inserting the clips, point to point testing as per the circuit diagram and plugging relays to Relay bases (except Inter-cables, relays other materials such as Rosine core, flux, nylon thread. PVC dotted tape should be supplied by the contractor). Point to point testing should be done in the presence of Engineer-in-charge as instructed in two stages, one before soldering and the other after soldering.	Numbers	960	174.00	167040.00
22	22	Fixing of approximately 200 nos of 6 way/ 1 way terminal block horizontally on 6mm Hylam sheets which in turn to be fixed to pre-erected relay rack with all supports as per instructions of SSE incharge. Materials such as bolts, nuts and supporting flats	Set	2	479.00	958.00

		to be supplied by the contractor only.				
23	23	Cutting and fixing of Hylam sheet of 6 mm/3mm thickness to suit to 1 way relay rack and fixing the following with contractors screws, bolts and nuts and fixing strips, condensers and resistances of various values - 50 No.	Lumpsum	5	103.00	515.00
24	24	Fixing of non-deteriorating fuse holders with LED indications and resistors and negative terminal strips on 6mm Hylam sheet with contractors screws, bolts and nuts.	Numbers	320	5.00	1600.00
25	25	Indoor Painting as per Railway standards. for 2/3/4/5/6 road stations.as per instructions of site incharge.	Lumpsum	2	3301.00	6602.00
26	26	Fabrication, fixing of required configuration of key in point machine for grouping of crank handle of point machines.	Numbers	3	82.00	246.00
27	27	Transportation of materials from any Stores to Site and vice-versa including loading and unloading - Lead up to 50kms. (Unit : PTPK)	MT-Km	1000	7.31	7310.00
28	28	Arranging of sufficient staff by contractor and giving test to the railway incharge for Stage -1 : testing of wires drawn point to point using buzzer. (Before and after soldering.)	Sheet	100	316.00	31600.00
29	29	Taking copiess of approved circuit diagrams/ drawings (A-2 size) issued by Railways on Ammonia Printing paper	Sheet	500	7.00	3500.00
30	30	Arranging rexin cloth bound booklets and binding for large size drawings (Unit : Each Booklet)	Each	4	174.00	696.00
31	31	Arranging rexin cloth bound booklets and binding for small size drawings (Each Booklet)	Each	4	142.00	568.00
32	32	Shifting of existing cables from old location boxes / relay room to new location boxes / new cable termination racks, terminating the conductors in serial order, bunching neatly for matching the circuits with new cables as per cable plan.	Per Conductor	700	8.00	5600.00

33	33	Carrying out circuit alterations in the existing location boxes as per the new circuits which require excavation and releasing unwanted cables from the location box and taking new cables inside the location, clamping of new cables, filling the location box with standard up to the bottom of location and plastering.	Numbers	10	612.00	6120.00
34	34	Erection of control panel - for way station with suitable foundation. The cables behind the panel shall be covered with mix of earth soil and sand. The top portion (floor) shall be plastered. If the tiles are removed for placing control panel, the tiles shall be re-fixed.	Numbers	2	5329.00	10658.00
35	35	Supply and fixing of different sizes of PVC casing /Pipes as required by Railway at site by fixing /clamping to the wall for indoor wiring purposes. . The material required carrying out the wiring such as soldering, lacing, casing, fixing clamps; screws etc are to be provided by contractor on his own cost.	RMT	50	32.00	1600.00
36	36	Earthing and soldering of cable armour as per instructions of site incharge. (Per Cable)	Each	300	21.00	6300.00
37	37	Numbering location box as per standard practice and writing all termination particulars inside the location box as per details. This includes all main and tail cable particulars, writing of cable numbering of cables, polishing teak wood boards in two coats and painting hard wood shelves in two coats of black enamel.	Numbers	30	280.00	8400.00
38	38	Arranging of sufficient staff by contractor and giving test to the railway incharge for Stage -II : Making a simulation test board plywood 3mm approximately size 1.5x1meter. Painted according to the yard layout fixing 1 way /2 way switches in its approximate position for track circuit , point setting lamp holders and lamps. Carrying out testing using simulation board after energising of internal circuit. Commissioning of indoor	Each	7	2686.00	18802.00

		and outdoor equipment as instructed by site incharge. (Unit : Per Road)				
39	39	Final location survey of cable route and final preparation of cable route plan as instructions of site incharge.	Route Kilo Meter (RKM)	3	994.00	2982.00
40	40	Breaking of metal/ tar road and refilling after laying the cables as per drawing no.SDO/CABLE LAYING/010. The surface of the road shall be concreted to match with the old surface	RMT	65	142.00	9230.00
41	41	Excavation of square coil pit up to a depth of 1 Mtr. X 1.5 Mtr. For providing coil at location box, point, signals etc., This includes back filling and ramming of trench / pit after laying the cable and consolidation of soil.	Numbers	1	181.00	181.00
42	42	Making a simulation test board (6 road means 2 Nos. 3 road stations) (Unit : Per Road)	Each	6	2686.00	16116.00
43	43	Cutting of rock to a depth of 200 mm and width of 200 mm where rock is visible from surface to a depth more than 300 mm up to 550mm, excavation of trench with width 300mm above rock concreting 200mmx200mm after laying of cable with 1 cement: 2 sand : 4 graded stone aggregate 20mm nominal size and curing of concrete to the required number of days, refilling with excavated soil ramming and consolidation except in bridges, culverts, level crossings, track crossings, cable pits, rocky soil including marking of cable alignment, clearing of debris etc., complete to the finished item of work as directed by site in-charge. Drg No.SK/CN/NEW/3. All the material is to be arranged by contractor	RMT	140	241.00	33740.00
44	44	When rock is visible for more than 0.5 metre and up to 0.8 metre from ground level, excavating the trench minimum 0.5metre and maximum 0.8metre, laying 6-quad cable as per the details shown in drawing no 3 and concreting 200mmx200mm over the rocky	RMT	140	200.00	28000.00

		surface after leveling with soil/sand. (Chase cutting not required).				
45	45	Supply & Laying of G.I Pipes 100mm dia. medium quality to specification IS.1239 (part-I) 1990 and galvanised to specification IS.4736-1986 On bridges , culverts, trench with channel ramps on either end of the culvert/bridges. Clamping arrangement with MS brackets for fixing of GI pipes on bridges/culverts, small holes of 5mm are to be drilled on the pipe before laying of the pipes.	RMT	180	341.00	61380.00
46	46	Foundations, erection, fixing of ' E' type locks, painting the box and painting of termination & other particulars of the Full location box . Metal Size of the aggregate should not be more than 40mm. Work to be done as per prescribed drawing no.SK/CN/1/89-A & B- All the material is to be arranged by the contractor except E- type lock (E-type lock will be supplied by Rlys)	Numbers	14	2460.00	34440.00
47	47	Excavation of earth and providing earth pipe of size 50mm dia, 3.5mm thick 2.0 m. long G.I. pipe with masonry tub constructed around at earth locations as specified by site Engineer. Alternate layers of 30kgs-chrcoal and 30kgs-common salt to be filled in the pit while closing. The pipe perforation etc., to be carried out as per drawing No. G/SG/09/2002. The earth pipe to be connected to the function with cable as supplied by Railways. The reading should be painted on the masonry tub. The masonry tub of 300X300X300mm should be covered with cement slab. (All materials to be supplied by the contractor except cable.)	Numbers	14	1406.00	19684.00
48	48	Test lamp arrangement in LBs/RE cuttings. (Auto mobile handle type) (Unit ; Per Location)	Each	12	119.00	1428.00
49	49	Straight through / derivation with/without transformer joints:	Each	28	979.00	27412.00

		Installation Thermo shrinkable jointing kits for straight through / derivation joints for LC gate, pump houses etc where VF transformers are not required and derivatin joint by using 2T trasformer (to be housed in the joint) . This includes tapping of nominated Quad at a specific location shown by the Railway Engineer at site to the jelly field cable already laid. All material including valco tapes except thermo shrinkable jointing kit shall be supplied by the contractor. (Refer Annexure for instructions on cable jointing). (Unit ; Joint)				
50	50	Provision and laying in position cement concrete of 1 cement:3 sand:6 graded stone aggregate of 20mm normal size excluding cost of cement, centering and shuttering (including transportation of all the material) .	cum	19	2085.21	39618.99
51	51	Cement 53 grade (@ the Rate (in Rupees) of 2.2 quintals per Cum of concrete including transportation	Quintal	41.8	460.00	19228.00
52	52	Transportation of materials from any Stores to Site and vice-versa including loading and unloading. -Lead above 50 Kms. (Unit : PTPK)	MT-Km	18000	4.80	86400.00
53	53	Excavation of trench to a depth of 1200mm and width of 300m at bottom in hard morum soil as per RDSO Sketch no SDO/CABLE LAYING /003 including marking of cable alignment, clearing of jungle, bushes, trial pits if required, refilling with excavated soil, ramming and consolidation after laying of cables complete to the finished item of work as directed by supervisor in-charge.	Route Kilo Meter (RKM)	0.5	67092.00	33546.00
54	54	Trenching under Rails for crossing the track(Track crossing) and refilling of trenches as per RDSO drawing No.SDO/CABLE LAYING /009. The track crossing shall be done at right angles.The depth of cable trench shall be 1.2 Mtr. below the rail flange. The depth between top cable and the level	RMT	50	141.00	7050.00

		of rail flange of the rail shall never be less than 1.0 Mtr.. The ballast shall be Stored separately while making the trench and the earth soil shall be stored separately. while re-filling the trench, the earth soil has to be filled first to the ground level and the ballast has to be properly put in the track. this work has to be done under supervision of supervisor of S&T and supervisor of P.Way				
55	55	Track crossing by horizontal boring method based on the drawing no.DRM/J/6735 dated 7-8-03 (enclosed) and as advised by the engineer at site. This includes the supply and insertion(pushing) of GI pipe (80mm) confirming to IS.1239 heavy class in the bore drilled with the coupler between two pipes. All the joints of casing pipes should be tight and leak proof.	RMT	15	1403.00	21045.00
56	56	Installation of single earth(maintenance free) including the following:- welding and fixing of copper tape (used as bus bar) welding of cable to copper plate using material supplied . All the material supplied for mounting of copper tape, exothermic weld materials, tools and fixtures for welding copper tapes shall be arranged by contractor.(As per bonding connections drawing no.SDO/RDSO/E&B/002 and as per RDSO spec no.RDSO/SPN/197 ver.1.0 or latest) a. Digging /Auguring/Boring of single earth pit of 100mm to 125 mm dia to the depth of approx. 2.8 meter. b. Insertion of 3.0 meter electrode and penetrating it at least .2 meter by pushing gently with sledge hammer where the natural soil is assumed. c. Filling of earth pit with earth enhancement compound (EEC) in slurry from and backfilling. d. Carrying out exothermic weld connection on single earth electrode to a copper tape of 150X25X6mm. e. Carrying out exothermic weld connection on	Set	2	4276.00	8552.00

		the copper tape of 150X25X6mm on earth electrode for connecting two 35 Sq mm cables for redundancy.. f. The exothermic welding material for bonding the earth system shall be procured from RDSO approved firms, which includes graphite moulds, clamps, weld metal powders. g. Digging a trench to a depth of at least 0 .5 Mtr. and run the two 35 Sq mm earth conductor (procured against supply item) from earth pit to MEEB. h. Routing and termination of two 35 sq mm earth conductors in trench from earth pit to 300X25X6mm MEEB, which is to be fixed inside the equipment room with two insulation studs. i. 525x350x250mm (inside dimension) concrete earth pit with a concrete lid, approx 50mm thick with pulling hooks, shall be provided to cover the pit. Back side of the cover will be painted in black and dare of the testing and average resistance value shall be written.				
					Total	2622317.39
Schedule	C-Schedule C					
Item - 3	Supply Portion Non SOR Items					
S No.	Item No	Description of Item	Unit	Qty	Rate	Amount
1	1	Supply of 6F Single mode optical fiber cable Make:Finolex/ Legrand / Systimax / panduit	RMT	1000	21.00	21000.00
2	2	Supply of Armoured 12 Core Fiber Cable, Make: Finolex or D-Link or BESTNET. Consignee Inspection.	RMT	2000	41.30	82600.00
3	3	Supply of 12 fiber single mode armoured optical fiber cable-multi loose tube out door direct burial Anti termite Anti rodent UV protected cable Make: R&M/Systimax/panduit	RMT	6000	87.00	522000.00
4	4	Supply of Multi Cord Binders SIZE 16 mm Make:MX or better (1.5 mts length) and as advised by supervisor in-charge.	Numbers	15	545.00	8175.00
5	5	Supply of Multi Cord Binders SIZE 22 mm Make:MX or better	Numbers	10	728.00	7280.00

		(1.5 mts length) and as advised by supervisor in-charge.				
6	6	Supply of Tie cable-Numbering Tags as specified by site incharge	Numbers	300	472.00	141600.00
7	7	Supply of Intelligent optical Multiplexure supporting E1 interface (E Cat) per specification given in Annexure - XVIII	Numbers	2	45725.00	91450.00
8	8	Supply of Micro phone AGN-500- or better (Make: Ahuja)	Numbers	5	1743.75	8718.75
9	9	Supply of Micro phone Stand Ahuja make(ATS-5) or better	Numbers	5	143.75	718.75
10	10	Supply of power supply unit (2 Amps) as per RDSO no IRS-TC 72/97 with amendment No.1 latest with 12v/7AH MF battery (To be procured from RDSO approved firms).	Numbers	6	3656.25	21937.50
11	11	Supply of 48 Core LIU, 19" Rack Mountable, with fully loaded SC/APC IO connector and Pigtaills Make: R&M / Systimax / panduit	Numbers	2	17582.00	35164.00
12	12	Supply of IP 55-15U/24U Height, Outdoor Street Cabinet with floor mount brackets including foundation, erection with proper protection of entry and exit cables as per site requirement and as per specification given in Annexure - XVIII	Numbers	1	32000.00	32000.00
13	13	Supply of Transmission Measuring Set Make:Anuvidyut Model No.368-C or better Make: Anuvidyut or better	Numbers	2	23010.00	46020.00
14	14	Supply of Multifunction office equipment as per specification given in Annexure - XVIII	Numbers	1	198900.00	198900.00
15	15	Supply of managed POE, 24 port Gigbit ethernet switch CBS 350-24P power budget 195W.	Numbers	2	38350.00	76700.00
16	16	Supply of 100 X 40 mm aluminium race ways of 2mm thick with all accessories	RMT	250	410.00	102500.00
17	17	Supply of 150x150 mm junction boxes with steel plate cover of 2mm thick for covering the tile as per the site requirement	Numbers	20	300.00	6000.00
18	18	Supply of CAT6 UTP cable for 10 baseT/100 BASE-TX / and 100	Metre	4575	20.00	91500.00

		BASE -T (GIGA BIT ethernet) make: D-Link/ HCL/Finolex				
19	19	supply of 4 way Spike busters with 5 mtrs power cable with individual switch And Fused for each socket 5A reputed make.NAX./ Similar	Numbers	8	1200.00	9600.00
20	20	Supply of RJ-45 Jacks and Termination with CAT-6 cable Make: D-Link.or similar	Numbers	900	10.75	9675.00
21	21	Supply of beetel GSM FWP 4G Fixed wireless telephone with all accessories.	Numbers	6	4307.00	25842.00
22	22	Supply of 24port CAT6 UTP cable patch Panel and as per specification given in Annexure - XVIII Make: Legrand / Systimax / Panduit or Similar	Numbers	3	7227.50	21682.50
23	23	Supply of Manageable L2 switch 12-Port 10/100/1000 POE+ BaseT with 2 Uplink optical 1G Ports upgradable to 2x10G SFP+ uplink ports similar to RUCKUS ICX7150 C12P or better Including all accessories licence, software up gradation and as per specification given in Annexure - XVIII	Numbers	3	28200.00	84600.00
24	24	Supply of Puncom make 2Mb loop protected programmable primary digital drop/ insert multiplexure with conference facility as per RDSO spec No. IRS-TC 68-2012 With Amend. 1 or latest, wired for 30 circuits complete with power supply, control and basic /common cards (ie PSU-2, TME-1, LPC-1) remote control supervision facility and all other accessories etc	Numbers	2	198712.00	397424.00
25	25	Supply of 2W/4W DTMF Head quarter control room Equipment as per RDSO specNo.IRS-TC 60/2007with amendment No.1 latest. (To be procured from RDSO approved firms)	Numbers	4	25960.00	103840.00
26	26	Supply of 10/100/1000 (1 GBPS) single mode media converter with accessories including connecting cables. Make: MRO-TEK/D-Link/ RAD/Team Engineer/Digisol or better as specified by Engineer Incharge.	Numbers	12	5752.50	69030.00

27	27	Supply of Amplifier 80W Ahuja make (SSB-80M) working 230V AC/12V DC	Numbers	4	5138.00	20552.00
28	28	Supply of printer Dot matrix compatable to existing data logger. Printer Dot matrix 80 column, 24 pin, 300 CPS equivalent to TVS-MSP 450 champion or better	Numbers	1	9606.00	9606.00
29	29	Supply of Voltage Stabilizer, capacity: 3 KVA, 1 Ph; Input Voltage 140-270V AC; Output voltage 230V (+/-) 10%, with a double pole switch of 32 Amps.	Numbers	17	13632.00	231744.00
30	30	Supply of 230V AC/12V DC SMPS charger with 10 Amps Make: Axion or better	Numbers	4	2150.00	8600.00
31	31	Supply of 12V/100 AH Batteries of make Exide/Amaron/HBL.	Numbers	15	8239.00	123585.00
					Total	2610044.50
Schedule	D-Schedule D					
Item - 4	Labour Portion Non SOR Items					
S No.	Item No	Description of Item	Unit	Qty	Rate	Amount
1	1	Supply and laying/ drawing the 20 Pair PVC switch board cable twisted in pairs PVC sheathed copper conductor 0.5mm dia (Make: Delton/Finolex/ Surabhi) through the PVC casing /PVC pipe (wall mounted/concealed) as specified by supervisor in-charge.	RMT	150	91.00	13650.00
2	2	Supply and laying/ drawing the 6 Pair PVC switch board cable twisted in pairs PVC sheathed copper conductor 0.5mm dia (Make: Delton/Finolex/ Surabhi) through the PVC casing /PVC pipe (wall mounted/concealed) as specified by supervisor in-charge.	RMT	150	36.00	5400.00
3	3	Supply and laying/ drawing the 10 Pair PVC switch board cable twisted in pairs PVC sheathed copper conductor 0.5mm dia (Make: Delton/Finolex/ Surabhi) through the PVC casing /PVC pipe (wall mounted/concealed) as specified by supervisor in-charge.	RMT	150	53.00	7950.00

4	4	Laying of U/G cable (PIJF Copper cable/OFC) on the outdoor structures such as over walls through pipes wherever required.For fixing the above, required fixing materials such as clamps, Screws,etc are to be arranged by the contractor.	Metre	100	100.00	10000.00
5	5	Supply and laying of DWC pipe made of polythene of size 103.5mm inner dia and 120mm outer dia in 6m length as per IS 14930 part-II along with 1 cuopler per pipe as per specification IP 67 grade for every 6 mts.with all accessories.	RMT	270	196.00	52920.00
6	6	Cutting of trench across/on the platform with cutting Machine as per site requirements for laying the cable through PVC/GI pipe and filling the trench after cable laying and restoring the surface with same/proper materials etc. and as per instructions of site in charge.	RMT	300	250.00	75000.00
7	7	Laying of 100X40mm aluminium race ways and fixing. Required couplings , clamps etc are to be arranged by the contractor	Metre	300	85.00	25500.00
8	8	Supply and fixing of PVC channels (Ganapati/Shreeram/star/similar make) with size of 1 mtr length, 100 mm x 50mm with top cover and laying the cables. Necessary fixing materials are to be arranged by the contractor. (3 anchor bolts per metre to be fixed),	RMT	20	595.00	11900.00
9	9	Supply and fixing of PVC channels (Ganapati/Shreeram/star/similar make) with size of 1 mtr length, 50 mm x 50mm with top cover and laying the cables. Necessary fixing materials are to be arranged by the contractor. (3 anchor bolts per metre to be fixed),	RMT	50	405.00	20250.00
10	10	Supply and fixing/laying of 25mm dia 2mm thick PVC pipe (Sudhakar/Nandi/SR make) duly grooving concealed in the walls. All the required accessories such as Deep boxes, Jn Boxes, Normal bends, diversion bends	RMT	400	78.00	31200.00

		etc are to be arranged by the contractor,				
11	11	Supply and fixing/laying of 40mm dia 2mm thick PVC pipe (Sudhakar/Nandi/SR make) duly grooving concealed in the walls. All the required accessories such as Deep boxes, Jn Boxes, Normal bends, diversion bends etc are to be arranged by the contractor,	RMT	150	169.00	25350.00
12	12	Supply & fixing of 25 mm PVC casing (Modi/Donaria Make) on the walls as required by Railways at site. Necessary fixing the above like screws, washers galvanized, wooden pegs teak etc are to be arranged by the contractor (5 screws per metre to be used)	RMT	100	41.00	4100.00
13	13	Supply & fixing of 40mm PVC casing (Modi/Donaria Make) on the walls as required by Railways at site. Hardware for fixing the above screws washers galvanized wooden pegs teak etc are to be arranged by the contractor (5 screws per metre to be used)	RMT	100	52.00	5200.00
14	14	Supply & installation of boxes with wago terminals (50 pair) of WAGO make	Numbers	2	5206.25	10412.50
15	15	Laying of PVC switch board cables /CAT6/ power /OFC 6F/12F cables in the PVC casing/PVC pipes as per the instruction of site incharge.	Metre	13000	5.00	65000.00
16	16	Supply And Installation of 42U x 800mm x 1000 mm with front perforated door and rear dual perforated door, side panels, 5/15 AMPs Vertical PDU -2 Nos, 4 X Fans, 2 x Shelf with earthing strip (copper)-1 with fasteners-2 pac, rack mountable DCDB panel with one common copper strip and 4 MCB -6A & 2MCB-16A. as per specification given in Annexure - XVIII	Numbers	3	60000.00	180000.00
17	17	supply and installation of Dual pole Wireless ADSL2+ 4 Port Modem Router with one Wan Port and RJ11 connector Make:Dlink,TPLink,Tenda	Numbers	14	1649.00	23086.00

18	18	Supply and fixing of 3U 19" rack mountable device to accommodate 13 I/O slots to populate Fast Ethernet / Gigabit Ethernet line cards with media conversions from single rack Make: MROTEK model no. OAK 13 / AC/R	Numbers	1	22700.00	22700.00
19	19	Supply and fixing of 2 channel Fast Ethernet media converter module with SC connector to be accommodated in OAK 13 rack Make: MROTEK / Similar	Numbers	4	4490.00	17960.00
20	20	Supply and fixing of 2 channel 10/100/1000 Ethernet media converter module with SC connector to be accommodated in OAK 13 rack Make: MROTEK /Similar.	Numbers	4	8075.00	32300.00
21	21	Supply and laying of SC-PC to SC-PC patch cords(5mts length) with 0dB adaptors as per TEC spec G/OJC-01/03 Jan 99 or latest as specified by supervisor in-charge Make:MX/D-Link/Bestnet/Dyeton	Pair	40	975.00	39000.00
22	22	supply and Laying of LC-LC to LC-LC patch cords(5Mtr) with 0dB adaptors as per TEC spec G/OJC-01/03 Jan 99 or latest specified by supervisor in-charge Make:MX/D-Link/Bestnet/Dyeton	Pair	10	1275.00	12750.00
23	23	Supply and Installation of USB 1TB external SSD Hard disk Make: Seagate/WD Elements or better	Numbers	1	7450.00	7450.00
24	24	Supply and Installation of CLIP Phones in each room similar to Model Beutel M-75 or latest	Numbers	15	1630.00	24450.00
25	25	Supply & fixing of 4 way type RJ-11 Jack suitable for telephones fixed on the modular PVC top covers including termination of cable pairs. Make Pointer/Roma or similar	Numbers	20	93.00	1860.00
26	26	Supply & fixing of 8 way RJ-45 Jack suitable for computer fixed on the modular PVC top covers including termination of cable pairs. Make: Pointer/Roma or similar	Numbers	25	400.00	10000.00
27	27	Supply & fixing of Dummies on PVC modular boxes 2/4/6/8	Numbers	50	25.00	1250.00

		modulars. Make Pointer/Roma or similar				
28	28	Supply & fixing of 2 modular PVC/MS box with top plate covers on the walls. Necessary fixing materials are to be arranged by the contractor. Make: Pointer/Roma or similar	Numbers	20	110.00	2200.00
29	29	Supply & fixing of 4 modular PVC/MS box with top plate covers on the walls Necessary fixing materials are to be arranged by the contractor. Make: / Roma. /Pointer or similar	Numbers	20	127.00	2540.00
30	30	Supply & fixing of 8/6 Modular PVC/MS box with top cover on the walls. Necessary fixing materials are to be arranged by the contractor. Make: Roma/ Pointer or similar	Numbers	20	137.00	2740.00
31	31	Supply & fixing of 9U rack with locking front glass door and fully vented rear steel door, compatible with standard 19" accessories, top cover with gland plate and fan, with rack cage nuts and screws, fan assembly kit, rack mount shelves, Power manager.etc this includes wiring as per the instruction of site incharge. Make: HCL/ Vero president/bestnet.	Numbers	7	8850.00	61950.00
32	32	supply & fixing of PVC Flexible pipe of 25/40/50mm dia sizes of reputed make	Metre	200	60.00	12000.00
33	33	Termination of 24Port UTP Patch panel as per instructions of Site Engineer (Fixing of Patch panel and Termination of CAT-6 Cables	Numbers	3	2100.00	6300.00
34	34	Supply and installation of 8 port unmanageable switch Make: Dlink/Tplink/Digisole	Numbers	9	1300.00	11700.00
35	35	Supply and installation of Rack for Testroom as per the specification given in Annexure - XVIII	Numbers	1	177000.00	177000.00
36	36	Installation of MUX 2Mb loop protected programmable primary digital drop/ insert multiplexure with conference facility as per instructions of site incharge	Numbers	2	18880.00	37760.00

37	37	Supply and installation of 4 wire DTMF way station equipment as per RDSO spec No. IRS - TC@45<47@2007 with amendment No.1 latest with 4 wire control phone as per spec no. IRS -TC@45&47@97 with amendement no,2 latest (To b procured from RDSO approvd firms)	Numbers	6	7080.00	42480.00
38	38	supply and installation of Digital clock with GPS sychronisation OFFICE CLOCK as per RDSO spec No.RDSO/SPN/TC @45>47@2008 REV.3.0 with amendment No.2 latest with RED LED To b procured from RDSO approved firms	Numbers	1	25960.00	25960.00
39	39	Supply and installation of multi service (chassis 1 U) Integrated access TDM Multiplexer with 4E1 interface up link (120Ohms balanced RJ 45 connectors) with 3 Module slots with AC/DC power supply model MAPLE - 4C Make: MROTEK or better	Numbers	7	35975.00	251825.00
40	40	Supply and installation of High speed providing 1/100 Mbps Fast Eathernet port , RJ45 connector suitable to accommodate in Mapple 4 slots Make: MROTEK /Puncum or better	Numbers	3	13847.50	41542.50
41	41	Supply of and installation of Audio interface card for user providing ten FXS links per card suitable to be accommodated in PCM multiservice integrated access TDM PCM multiplexer. suitable for Maple 4C. Make: MROTEK/Cygnus	Numbers	7	26600.00	186200.00
42	42	Supply of and installation of Audio interface card for user providing ten FXO links per card suitable to be accommodated in PCM multiservice integrated access TDM PCM multiplexer.	Numbers	3	29312.50	87937.50
43	43	Supply and installation of 2W/4W E&M interface module with eight interfaces suitable for Maple 4C. Make: MROTEK/Cygnus	Numbers	7	23700.00	165900.00
44	44	supply and installation of 1U link panel set	Numbers	1	24340.00	24340.00
45	45	Supply and installation of SIP Basic Phones.	Numbers	12	4767.20	57206.40

46	46	Supply & installation of FRFC SMPS Based power plant 48V, 50 Amps capacity SMPS charger with 48V/25Amps SMR module 3-Nos (2 X 25Amps+1 X 25Amps)) suitable for VRLA battery as per RDSO Spen No. RDSO/SPN/TL/ 23/99 (Ver.4) or latest for Telecom purpose.	Numbers	14	120024.00	1680336.00
47	47	Supply & installation of VRLA battery bank 48V. 2V X 24 Nos. (300 Ah capacity) as per RDSO Specn No. IRS.S.93/96 (A) (Amendment 1) or latest.	Set	2	93765.00	187530.00
48	48	Supply and installation of Class-B & C Type Lightning Cum surge protection device with Visual indication, pluggable and testable with surge tester and potential free contacts. Outputs suitable for 230V AC applications (Conforming to RDSO Specification No.RDSO/SPN/165/2012 version.3 or latest and should be tested as per IEC 61643). it also includes supply and installation of Surge Protection devices duly wired mounted on din rail with necessary fittings like Nuts and bolts etc. Quantity of each type will be decided by the Engineer in charge as per the site requirements.	Numbers	1	24972.50	24972.50
49	49	Supply and installation of 10KVA true online ups system (FMI-with iso transformer),batteries SMF 12 V / 42AH-20 Nos, battery stand and interlink cables. Make: Aplab/ Emmerson /Numeric/APC/ Libert. or better	Numbers	1	208500.00	208500.00
50	50	Supply & fixing of 20/10A 5 Pin twin socket heavy duty with safety of reputed make	Set	195	141.00	27495.00
51	51	Supply & erection of 20A one way switch of reputed make	Each	10	88.00	880.00
52	52	Supply and fixing of MCB switch BOX with one 32 Amps and termination of AC wires from SPD and at input of charger .make: Legrand or better	Numbers	10	1200.00	12000.00
53	53	Supply and Installation of LED Display units 55" Make : samsung\LG	Numbers	2	118000.00	236000.00

54	54	Supply and laying of HDMI Cable 20 Mtrs length with (Male-Female/Female-Female/Male-Male) moulded type as required at site.Make:MX/ ATEN/ Dyeton	Numbers	2	3800.00	7600.00
55	55	Fiber splicing and termination in LIU at Central office, street cabinets, subscriber end	Pair	200	88.50	17700.00
56	56	Supply and installation of control console unit as per specification given in Annexure - XVIII	Numbers	5	50000.00	250000.00
57	57	Installation Labour, Design and Project Management, Programming, Operation and Maintenance Manuals - including Equipment Manuals & Training - Operation and Maintenance. Provision of ring earth with min 03 MFE earths & below 1 ohm earthing and connectivity to Equipment	Lumpsum	1	218300.00	218300.00
58	58	Supply and laying of CAT-6 cable patch cards-2mtrs length with RJ-45 Jacks on both ends Make: D-Link/Bestnet	Numbers	20	180.00	3600.00
59	59	Supply and installation of 5KVA UPS with battery backup as per the specification given in Annexure - XVIII	Numbers	1	141600.00	141600.00
60	60	Supply & installation of 2KVA UPS systems with battery back up of 120 mins Make: APC/Numeric/Emerson with Battery minimum 65 AH each	Numbers	1	57907.00	57907.00
61	61	Supply & installation of 24 port IPDSLAM Model : VES 1724-56 along with Telco 50 pair cable-02 Nos of Make ZYXEL as per specification enclosed or advanced version of equivalent make. as per the specification given in Annexure - XVIII	Numbers	1	91981.00	91981.00
62	62	Supply and Installation of PVC Conduits/ Casing Capping of defferent sizes as per site requirement (All required materials are to be arranged by the contractor)	Metre	300	36.29	10887.00
63	63	Supply and installation of Wi-Fi access point with POE facility Make: TP link/D-Link.	Numbers	10	4715.00	47150.00
64	64	Fabrication and Supply of Powder coated Aluminum ladder of Size 38mmX38mmX5mm and	RMT	30	518.00	15540.00

		250 mm wide , with 38mm X5mm flat fixed at intervals of not more than 250 mm. All nuts, bolts, PVC sleeves, insulators etc. are to be arranged by the contractor.				
65	65	Installation,testing and commissioning of Electricallly operated lifting barrriers with assembly, drive mechanism, motors etc. with control panel, boom locking arranmgements & crank handles complete with Hand generator as per detailed technical specification and instructions of site in-charge (The above work includeds excation of pits & concreting foundation for pedestal - 2 nos, ecavation of pits & concreting foundation for gate meeting post-2 nos, Erection of Electric lifting barriers-2 set, fixinig of ELB panel & it's wiring -1 no as per detailed technical specification and instructions of site in charge.	Set	1	30000.00	30000.00
66	66	Supply of teak wood board for fixing 2 Nos of EKTs, with 2 Nos of push button, 6 nos 10 mm LED resistance holders etc. (only EKTs will be supplied by Railways)	Set	1	5725.00	5725.00
67	67	Releasing of relay racks complete with its wiring, relay base, supporting frames, cable ladder, indoor/outdoor cables etc. Axle counter wiring, Data logger and its wiring, panel and its wiring completely and any of the miscellaneous wiring in relay/equipment/panel room etc. complete.	Lumpsum	1	24882.00	24882.00
68	68	Supply and binding of Album type booklet with C- size plastic transparent covers of approved quality.	Numbers	5	590.00	2950.00
69	69	Supply and installation of wago connectors FTB set (per unit means per location)	Numbers	8	2000.00	16000.00
70	70	Supply and installation of wago connectors UFTB set (Un-fused terminal block) (Unit : Per location)	Numbers	8	1800.00	14400.00
71	71	Supply and installation of wago connector End Plate set (End	Numbers	8	1300.00	10400.00

		Clamp) (Unit ; Per Location)				
72	72	Supply and installation of Din Rail (Unit : Per Location)	Numbers	8	350.00	2800.00
73	73	Supply, installation, testing and commissioning of 24 port Gigabit with 2 nos of loaded 1G SFP modules PoE+ switch(L-2 Switch) with 370W or higher	Numbers	4	59000.00	236000.00
74	74	Network video Recorder (NVR): Supply, installation, integration and commissioning of ONVIF 32 channel NVR of similar to model No.4432-4KS2 or latest of Dahua Make or better /Hikvision/CP plus and as per specification under • Main Processor: Quad-core embedded processor • Interface: 1 HDMI, 1 VGA, • Resolution: 1920 x 1080;• Multi-screen Display:1/4/8/9/16/32 • OSD: Camera title, Time, Video loss, Camera lock, Motion detection, Recording; • Trigger Events; Recording, PTZ, Tour, Alarm Out, Video Push, Email, FTP, Snapshot, Buzzer and Screen Tips; Video Detection: Motion Detection, MD Zones: 396 (22 x 18),Video Loss and Camera Blank; • Playback:32; • Search Mode: Time /Date, Alarm, MD and Exact Search (accurate to second), Smart search; • Playback Function: Play, Pause, Stop, Rewind, Fast play, Slow Play,Next File, Previous File, Next Camera,Previous Camera, Full Screen, Repeat, Shuffle, Backup Selection, Digital Zoom; • Backup Mode: USB Device/Network/ Internal SATA burner/eSATA Device; • Internal HDD: 4 SATA III Ports, Up to 6 TB capacity for each HDD;• USB: 2 USB Ports (1 USB 2.0 , 1 USB 3.0); • RS232: 1 Port, for PC Communication and wireless Keyboard & optical mouse	Numbers	2	44000.00	88000.00
75	75	External Hard Disc:: Supply of Expansion desktop 10TB external Hard drive HDD USB 3.0 for PC/Laptop. Make: Model Seagate(STEB10000400) or latest or WD/ADATA make	Numbers	4	24000.02	96000.08
76	76	Supply & Installation 5MP or higher resolution Onvif POE Type	Numbers	24	11200.00	268800.00

		IP Bullet Camera with all accessories. Make: Hikvision/Axis/Sony/Dahua Model DS-2CD2T55FWD-I5(4mm) or latest. as per the specification given in Annexure - XVIII				
77	77	Supply and installation of 8 port switch with POE (power over Ethernet) 8 port PoE switch shall be of following configuration. 8 x 10/100/1000BASE-T PoE ports. Supports IEEE 802.3at PoE Compact design for convenient placement Fanless design for silent operation Innovative Green energy saving features Complete web interface for comprehensive switch management. These switches shall be compatible and seamlessly integrated with existing CCTV network Make: Netgear/Cisco/Juniper or similar	Numbers	6	7965.00	47790.00
78	78	Supply of Fibershot 32 decibel Single Mode Optical Time Domain Reflectometer-(OTDR) as per the specification given in Annexure - XVIII	Numbers	1	73800.00	73800.00
79	79	Supply of Auto single fibre splicer (Splicing machine) with all accessories like stripper, cleaver etc as per the specification given in Annexure - XVIII	Set	1	99998.00	99998.00
80	80	Preparation of auto cad drawings in A-2 size such as external/internal wiring particulars, contact analysis, P & K Rack particulars and datalogger particulars ect. as required at site and instructions of site incharge. 1 set of prints on tracing film and a soft copy on compact disc (CD) to be supplied	Sheet	50	600.00	30000.00
81	81	Excavation of pit and casting concrete foundation as per Drg. No. SC/N/CN / 62 /B/1 using metallic templates, for erection of GROUND TYPE Shunt signal. (Foundation bolts and nuts, cement, river sand, stone jelly of size 20/25mm and allother miscellaneous materials for the work shall be supplied by the contractor).	Numbers	1	2519.00	2519.00

82	82	Erection of Ground type shunt signal complete including surface base, signal pole, LED aspects/Lenses, bulbs with holders (whichever is applicable), fixing of lens guards, number plate, termination of tail cables, wiring of signal unit with PVC wire 3/0.75mm copper, provision of EWS locks and painting of one coat of redoxide and two coats of Aluminium /.enamel paint, (Supply of Ground type shunt signal complete including surface base, signal pole, LED aspects, holder, bulbs and lenses is not covered in this schedule). [Wire PVC 3/0.75mm copper, lens guards, EWS locks, enamelled number plates, bolts and nuts, lead wool, paints and all other miscellaneous materials shall be supplied by the contractor).	Numbers	1	2097.00	2097.00
83	83	Excavation of pit, casting concrete foundation and erection of apparatus case full size as per Drg. No. SK/CN/1/89/A/1 and fixing of 1 No. of 'E' type lock for the front door, fixing of one hylam sheet of 1000mm X 940mm X 10mm thickness and painting the apparatus case inside and outside with one coat of redoxide and two coats of aluminum paints (foundation bolts and nuts, 'E' type lock, cement, river sand, stone jelly of size 20/20mm, hylam sheet of size 1000mm X 940mm X 10mm, paints, varnish, fixing bolts and nuts and all other miscellaneous material required for the work shall be supplied by the contractor).	Numbers	1	9875.00	9875.00
84	84	Fabrication & supply and fixing of stay rods for wiring support at location boxes, CT rack etc., this includes providing of sleeve to stay rod, fixing bolts, nuts & washers etc as required at site.	Numbers	100	265.00	26500.00
85	85	Supply, Installation, Testing and Commissioning of 1 Loop Addressable fire alarm Panel which consists of display board which will have 160 Character Display board with Touch Key pad. Mother board which will	Numbers	1	187500.00	187500.00

		have CORTEX 32 Bit Processor, Inbuilt RS-485 Networking Circuit for peer to peer networking, Inbuilt USB 2.0 Port for upload and download the configuration tools, two inbuilt programmable inputs, two inbuilt NAC's and Three programmable relays and 4 Loop card circuits, power supply board which will have SMPS fully protected board, which its operates on 120 to 220V AC +/- 10% with 50/60 Hz, battery backup 24V DC with built-in charger. The panels have IP-50 rating optional: for BMS integration MODBUS converter, Graphical Software, Software for TCP/IP module, provision for Auto dialer approval UL listed. As per RDSO Specification No RDSO/SPN/217/2018 ver 2.0 or latest(Inspection by RDSO)				
86	86	Supply, Installation, Testing and Commissioning of Addressable Multi detector combination of heat cum smoke, with isolator base binary coded address setting. Dual LED'S Tor 360 degrees visibility, Sleek low-profile housing design, Regular 100mm base. IP Rating: IP-42. Approval: UL listed, (As per RDSO Specification No. RDSO/SPN/217/2018 ver 2.0 or latest.(Inspection by RDSO)	Numbers	4	6750.00	27000.00
87	87	Supply, Installation, Testing and Commissioning of Sounder cum strobe 24V operated 85db sounder output, 75Cd flasher controlled through addressable control module (As per RDSO Specification No. RDSO/SPN/217/2012 ver 2.0 or latest)(Inspection by RDSO)	Numbers	1	7500.00	7500.00
88	88	Supply. Installation, Testing and Commissioning of 2C X 1.5 sqmm unarmored fire survival cable (As per RDSO specification No. RDSO/SPN/217/2018 ver 2.0 or latest)(Inspection by RDSO)	Metre	100	300.00	30000.00
89	89	Supply and installation of Multifunction MachinesMfm, (EPSON L15160) as per specification given in Annexure - XVIII	Numbers	1	60000.00	60000.00

90	90	Replacement of print head for existing HP design jet plotter as per specifications given in annexure - XVIII	Numbers	1	42500.00	42500.00
91	91	Supply and installation of MS Powder Coated and Plastic moulded seat as per specifications given in annexure - XVIII	Numbers	90	1475.00	132750.00
92	92	Supply and placing of trucking and capping (RCC Duct) of 500mm width as per Drg No. SDO/Cable laying/007 (Annexure-10) as Detailed in document No. RDSO /SI/G/ 2010 Ver1.1 or Latest on Guidelines on signalling cable laying. (Inspection by RITES).	Numbers	120	1278.00	153360.00
93	93	Excavation of cable trench in all kind of soil except hard rocky areas in cluding clearing of roots of tree rocks, bushes etc., to depth of 0.5 mtr-0.6 mtr to accommodate the R.C.C duct and to a width of 0.5 mtr. This shall include refilling of earth as per instructions of Engineer incharge. Laying of cables is not included in this schedule in case of less depth payment will be done proportionate to depth subjected to minimum depth of 500 mm	Metre	84	172.50	14490.00
94	94	Supply of 2 Wire 2Mbps G.SHDSL modem consisting of one E1 unit and one V.35 units with builtin power supply AC or DC from single socket which supports pluggable DTE interface card to accommodate G.703 E1, V.35, ETH, EL to V.35 converter, E1 to ETH converter. The unit to have provision to interoperable with a ceter located rack with SHDSL modules (tec approved interfaces E1, V.35, E1 to V.35 and Ethernet) Model : WL/E1/2W/AC/DC, MAKE : mrotek/ actel/ gygnus	Pair	1	29780.00	29780.00
95	95	Supply of 10/100 Mbps single mode media converter with accessories including connecting cables. Make: MRO-TEK/D-Link/RAD/Team Engineer or better as specified by Engineer Incharge.	Numbers	4	4000.00	16000.00

96	96	Supply & installation of 108 Mbps TDM over Ethernet Access Multiplexer with provision of four DTE E1 channel over fiber port, fiber ports supporting single mode with driving distance upto 40kms. Plus one Ethernet Port 10/100 Mbps, Power supply AC & DC option from single socket.	Numbers	2	28320.00	56640.00
97	97	Supply and installation of Calss D/C type Lightning Cum surge protection safety device with Visual indication, pluggable and testable with surge tester and potential free contacts. Outputs suitable for 48V DC applications (Conforming to RDSO Specn No. RDSO/ SPN/165/2012 or latest and should be tested as per IEC 61643) it also includes supply and installation of Surge Protection devices duly wired mounted on din rail with necessary fittings like Nuts and bolts etc. as per the site requirements and the Engineer in charge.	Set	4	5236.00	20944.00
98	98	Supply and installation of 3KVA UPS with 2 Hours backup make APC model SRCE3KUXI or better with 65 AH12v exide brand SMF batteries(6nos) with battery links . Make: Aplab/ Emmerson /Numeric/APC/Libert.)	Numbers	1	105020.00	105020.00
99	99	Supply and installation of 1 KVA UPS with 30 minutes backup Make: Aplab/ Emmerson /Numeric/APC/Libert.)	Numbers	1	7000.00	7000.00
					Total	7153218.48
Schedule	E-Schedule E					
Item - 5	SUPPLY PORTION (STMs)					
S No.	Item No	Description of Item	Unit	Qty	Rate	Amount
1	1	Supply of managed expandable 19" RACK mountable 48V DC working add-drop STM-16 SDH equipment complete with redundant power supply cross connect module (20G) with 2 nos of STM-16 of aggregate L16.2 optical interface modules/cards one each for east and west side TX/RX complete with 10m long optical patch cards and SFPs and 4 nos of	Numbers	2	708000.00	1416000.00

		STM-4 S1.1 optical interface module complete with 10m long optical patch cards and SFPs and 2 nos of STM-1 S 4.1, software license for STM 4/16, 63/21 EIs card Ethernet febric card with 2 x gigE+4X FE, order wire filter unit, FAN Unit, subrack DDF installation material manuals and other accessories complete, engineering order wire (EoW), Necessary connecting cards cable for power supply modules with basic and enhanced installation kit as per the TEC specification with latest amendment it should be compatible and synchronize with existing link and manageable with existing NMS. as per the specification given in Annexure - XVIII				
2	2	Supply and installation of STM-1 equipment with all accessories and as per specification given in Annexure - XVIII	Numbers	16	296600.00	4745600.00
3	3	Supply and installation of STM-4 equipment with all accessories and as per specification given in Annexure - XVIII	Numbers	6	595800.00	3574800.00
					Total	9736400.00